

CALVIN YI-REN CHEN

Curriculum Vitae

8N2, Cosmology Hall
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🌐 [My Webpage](#)

Education and Employment

- 2024– **Postdoctoral fellow**, *Leung Center for Cosmology and Particle Astrophysics & Max Planck-IAS-NTU Center*, Taiwan.
- 2020–2024 **Doctor of Philosophy, Theoretical Physics**, *Imperial College London*, United Kingdom.
Imperial College London President's PhD Scholarship
- 2019–2020 **Master of Advanced Study, Applied Mathematics**, *University of Cambridge*, United Kingdom.
- 2016–2019 **Bachelor of Science, Physics with Theoretical Physics**, *Imperial College London*, United Kingdom.

Publications

- Apr 2026 Manifest duality and Lorentz covariance for linearised gravity as edge modes. Calvin Y.-R. Chen, Euihun Joung, and Karapet Mkrtchyan.
- Jan 2026 Holomorphic structure of massive scalar fields in $(A)dS_2$. Calvin Y.-R. Chen, Lukas W. Lindwasser, and Massimo Porrati.
- Sep 2025 Causality in the presence of stacked shockwaves. Calvin Y.-R. Chen, Aoibheann Margalit, Claudia de Rham, and Andrew J. Tolley. *Phys. Rev. D*, volume 111, page 024066.
- Aug 2025 Deformations of extremal black holes and the UV. Calvin Y.-R. Chen, Claudia de Rham, and Andrew J. Tolley. *Phys. Rev. D*, volume 111, page 024056.
- Jul 2025 On the Aretakis instability of extremal black branes. Calvin Y. R. Chen and Áron D. Kovács. *JHEP*, volume 09, page 084.
- Jan 2025 Higher-order chiral scalar from boundary reduction of 3d higher-spin gravity. Calvin Y.-R. Chen, Euihun Joung, Karapet Mkrtchyan, and Junggi Yoon. *JHEP*, volume 05, page 186.
- Dec 2022 A cautionary case of casual causality. Calvin Y.-R. Chen, Claudia de Rham, Aoibheann Margalit, and Andrew J. Tolley. *JHEP*, volume 03, page 025.
- Jan 2020 Quantifying athermality and quantum induced deviations from classical fluctuation relations. Zoe Holmes, Erick Hinds Mingo, Calvin Y.-R. Chen, and Florian Mintert.

Talks (Invited)

Conferences

- Apr 2026 Gravity2026, IBS Daejeon
- Feb 2026 Quantum Gravity and Cosmology, Bologna
- Jan 2026 JGRG34, Kyoto
- Jul 2025 GR24/Amaldi16, Glasgow
- Jun 2025 GbRP25, Sogang University
- May 2024 UKCosmo, King's College London
- Apr 2024 BritGrav24, Queen Mary U London
- Jun 2022 Gravity, YITP Kyoto University
- May 2022 SCGSC 2022, University of Amsterdam

Seminars

Sep 2025 HEP seminar, National Yang Ming Chiao Tung University
Mar 2025 GAnG seminar, Queen Mary University London
Dec 2024 Research seminar, ITMP Moscow State University
Jul 2024 Colloquium, Imperial College London
Feb 2024 Cosmo Donuts, Imperial College London
Feb 2024 String Phenomenology (Online Seminar Series)
Dec 2023 Theory group PhD student seminar, Imperial College London
Nov 2023 Astrophysics seminar, YITP Kyoto University
Nov 2023 Group seminar, Kindai University
Oct 2023 Research seminar, ITMP Moscow State University
Oct 2023 Theory seminar, Carnegie Mellon University
Sep 2023 Particle theory seminar, Case Western Reserve University
Aug 2023 Group seminar, National Taiwan University
May 2023 GR seminar, University of Cambridge
Nov 2022 Theory group PhD student seminar, Imperial College London
Oct 2022 Copernicus Webinar series
May 2022 Theory group PhD student seminar, Imperial College London
Feb 2022 Theory group PhD student seminar, Imperial College London
Nov 2021 Theory group PhD student seminar, Imperial College London
Nov 2020 Non-equilibrium systems group meeting, Imperial College London

Teaching

Jun 2024 **Lecturer**, "*AdS₂ Holography*" *Special Topics Classes*, MSc QFFF, Imperial College London.
Jun 2023 **Lecturer**, "*JT Gravity*" *Special Topics Classes*, MSc QFFF, Imperial College London.
2022 **Teaching assistant**, *Advanced Classical Physics*, Y3 UG, Imperial College London.
2021–2023 **Teaching assistant**, *Advanced Quantum Field Theory*, MSc QFFF, Imperial College London.
2020–2021 **Teaching assistant**, *Quantum Mechanics*, Y2 UG, Imperial College London.

Professional Experience

2025– **Organiser**, *LeCosPA Seminar*, Leung Center for Cosmology and Particle Astrophysics.
2024– **Organiser**, *Journal Club*, Leung Center for Cosmology and Particle Astrophysics.
Apr 2023 **Organiser**, *Intersections of String Theory and QFT III*, Imperial College London.
Jan 2023 **Organiser**, *UK-QFT XI*, University of Cambridge.
2022–2024 **Organiser**, *Theory Group Journal Club*, Imperial College London.

Honours & Awards

2022 Best poster at Gravity@Prague
2022 Dean's Fund
2016 German Physical Society Abitur Prize
2016 German Mathematical Society Abitur Prize

Outreach

- Jul 2024 **Volunteer**, *Year-12 Visit*, Imperial College London.
Jun 2024 **Volunteer**, *Great Exhibition Road Festival*, Imperial College London.
Jan 2024 **Volunteer**, *Abdus Salam Celebration*, Imperial College London.
Faculty Natural Sciences Prize for Excellence
2023 **Mentor**, *Imperial College London Maths School*.
2019 **Volunteer**, *Mexican Caribbean Manta Project*.

Languages

native German
fluent English
fluent Mandarin

References

Andrew J. Tolley

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